

FRESNO YOSEMITE, CA, USA

WMO: 723890

Lat: 36.780N

Lon: 119.719W

Elev: 333

StdP: 14.52

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 93193

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	32.5	34.8	22.9	17.4	58.8	26.4	20.6	56.2	18.5	52.7	16.5	52.2	2.0	90	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.1	103.8	69.6	101.3	68.6	98.8	67.7	73.1	96.9	71.4	94.6	69.9	92.9	8.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	94.0	84.7	62.8	86.6	83.5	60.8	80.6	81.5	36.8	96.6	35.3	94.7	34.0	92.8	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.4	14.4	DB	29.3	108.2	2.6	1.9	27.5	109.6	26.0	110.7	24.5	111.8	22.7	113.2	(4)
(5)			WB	28.1	75.6	2.6	3.6	26.2	78.2	24.7	80.3	23.2	82.4	21.3	85.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	48.6	52.2	57.7	62.4	70.0	77.8	83.8	82.4	77.1	66.5	55.5	47.9	(6)	
	DBStd	13.80	5.00	4.92	5.93	7.01	7.11	6.80	5.11	4.68	6.06	6.04	5.87	5.11	(7)	
	HDD50	246	88	30	7	1	0	0	0	0	0	0	18	102	(8)	
	HDD65	2138	510	358	235	131	30	2	0	0	2	53	288	529	(9)	
	CDD50	5807	44	92	247	372	621	834	1048	1004	812	512	183	38	(10)	
	CDD65	2223	0	0	9	52	186	387	583	539	364	100	3	0	(11)	
	CDH74	29602	0	4	162	710	2381	5265	8270	7237	4414	1103	56	0	(12)	
	CDH80	16698	0	0	26	244	1094	3005	5161	4397	2391	375	5	0	(13)	
(14) Wind	WSAvg	5.9	4.0	4.8	5.8	7.2	8.2	8.4	7.5	6.8	5.9	4.6	3.7	3.8	(14)	
(15) Precipitation	PrecAvg	11.0	2.1	2.0	2.0	0.9	0.4	0.2	0.0	0.0	0.2	0.6	1.3	1.5	(15)	
	PrecMax	21.6	8.5	6.1	7.2	4.4	1.7	1.9	0.2	0.2	1.2	2.4	3.5	4.3	(16)	
	PrecMin	6.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	3.9	1.7	1.7	1.6	1.0	0.5	0.4	0.0	0.0	0.4	0.7	1.0	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.3	73.9	83.5	91.7	99.3	105.3	107.4	106.3	102.5	94.0	79.7	67.0	(19)	
		MCWB	53.0	57.3	60.5	62.8	66.8	69.9	70.7	70.8	68.1	65.5	59.6	55.0	(20)	
	2%	DB	64.0	69.2	78.4	86.6	94.6	101.6	104.2	102.7	98.9	88.0	74.9	63.0	(21)	
		MCWB	53.8	56.9	59.9	61.7	65.4	68.4	69.9	69.2	67.3	63.3	58.3	53.3	(22)	
	5%	DB	60.6	65.5	74.6	82.1	90.5	98.1	101.6	100.1	96.0	84.0	70.6	60.0	(23)	
		MCWB	52.7	54.7	58.2	60.2	63.6	66.8	69.1	67.9	66.5	61.7	57.0	52.0	(24)	
	10%	DB	57.7	62.2	70.8	77.4	86.3	94.5	98.8	97.4	92.7	80.4	66.8	57.1	(25)	
		MCWB	51.8	53.5	57.0	58.9	62.2	65.8	68.4	67.4	65.6	60.4	55.5	51.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.9	60.5	64.3	66.6	70.0	74.1	75.4	76.4	72.3	67.9	62.4	58.9	(27)	
		MCDB	63.2	67.7	77.5	82.9	92.6	98.3	100.3	98.7	94.1	86.1	74.4	63.6	(28)	
	2%	WB	55.9	58.3	62.0	63.6	67.6	71.7	73.3	72.5	70.2	65.8	60.2	55.9	(29)	
		MCDB	60.0	66.2	74.8	81.3	90.4	95.4	98.3	94.7	91.5	83.2	71.1	59.6	(30)	
	5%	WB	54.1	56.4	59.8	61.6	65.5	69.4	71.5	70.5	68.3	63.7	58.5	53.9	(31)	
		MCDB	58.3	63.6	71.4	78.8	86.6	92.0	95.4	93.6	90.5	80.5	68.1	57.7	(32)	
	10%	WB	52.2	54.5	57.9	59.7	63.4	67.6	70.1	68.9	66.9	61.9	56.6	51.9	(33)	
		MCDB	56.7	60.8	68.4	75.3	83.1	90.1	93.6	92.2	89.2	77.4	64.8	55.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	18.1	21.2	23.3	25.6	28.4	29.1	29.4	27.7	24.5	19.8	16.1	(35)	
		MCDBR	21.3	23.2	27.4	30.0	30.6	30.9	30.0	30.8	30.7	29.4	25.9	20.3	(36)	
		MCWBR	12.5	12.2	11.6	10.6	9.9	9.2	9.0	8.8	9.2	10.5	12.1	11.8	(37)	
	5% WB	MCDBR	15.8	20.2	24.7	27.3	28.9	29.4	28.3	29.2	28.5	26.5	22.4	13.7	(38)	
		MCWBR	10.1	11.3	11.6	10.4	10.0	9.6	9.7	9.5	9.2	9.8	11.3	9.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.359	0.372	0.395	0.378	0.368	0.364	0.383	0.368	0.371	0.390	0.380	0.371	(40)	
	taud		2.324	2.304	2.239	2.331	2.387	2.399	2.374	2.389	2.391	2.283	2.253	2.266	(41)	
	Ebn at Noon		259	271	275	286	290	290	283	286	279	260	247	243	(42)	
	Edh at Noon		31	35	41	40	38	37	38	37	35	36	33	30	(43)	
(44) All-Sky Solar Radiation	RadAvg		773	1105	1572	2001	2379	2669	2570	2357	1958	1422	962	705	(44)	
	RadStd		111	129	136	152	137	98	97	72	60	81	54	98	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.01	N/A	-2.20	N/A	N/A	-1.44	-68	-223	+344	+205	(46)
(47)	Regional (23 neighbors)	+0.92	N/A	-2.05	+0.79	N/A	N/A	N/A	-245	+301	+95	(47)

Nomenclature: See separate page